

Project Title

Microsoft Entra ID Identity and Access Management (IAM) with Zero Trust Security Lab

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Project Report

1. Executive Summary

Organizations increasingly rely on cloud-based identity management to secure access to applications, data, and digital resources. Traditional security models assume that users inside the corporate network are trusted; however, this assumption is no longer sufficient in today's threat landscape. Modern organizations adopt the Zero Trust security model, where every user, device, and request must be verified before access is granted.

This project demonstrates the implementation of an Identity and Access Management (IAM) laboratory using Microsoft Entra ID (formerly Azure Active Directory). The environment simulates a small organization called CyberSecurity Lab by creating users, security groups, administrative roles, and role-based access controls. The project also documents Joiner-Mover-Leaver (JML) processes, access reviews, and security policies to represent enterprise identity management practices.

Although advanced features such as Conditional Access and enterprise Multi-Factor Authentication require Microsoft Entra ID Premium licensing, the project successfully demonstrates the fundamental concepts of IAM, RBAC, and Zero Trust using the Microsoft Entra ID Free edition.

2. Introduction

Identity is the new security perimeter. As organizations migrate applications and services to the cloud, protecting user identities becomes increasingly important. Microsoft Entra ID provides centralized identity management, authentication, authorization, and access governance capabilities that help organizations secure their environments.

This project focuses on implementing a practical Microsoft Entra ID laboratory where users, groups, administrative roles, and security policies are configured according to cybersecurity best practices.

3. Project Objectives

The primary objectives of this project are:

- Deploy a Microsoft Entra ID tenant.
- Create organizational users.
- Create security groups.
- Assign users to departments.
- Implement Role-Based Access Control (RBAC).
- Apply the Principle of Least Privilege.
- Document Joiner-Mover-Leaver workflows.
- Perform access reviews.
- Create organizational security policies.
- Prepare project documentation suitable for GitHub and professional portfolios.

4. Problem Statement

Many organizations struggle with identity sprawl, excessive permissions, weak authentication, and poor user lifecycle management. These issues increase the risk of unauthorized access, insider threats, and data breaches.

The project addresses these challenges by implementing centralized identity management using Microsoft Entra ID while following Zero Trust security principles.

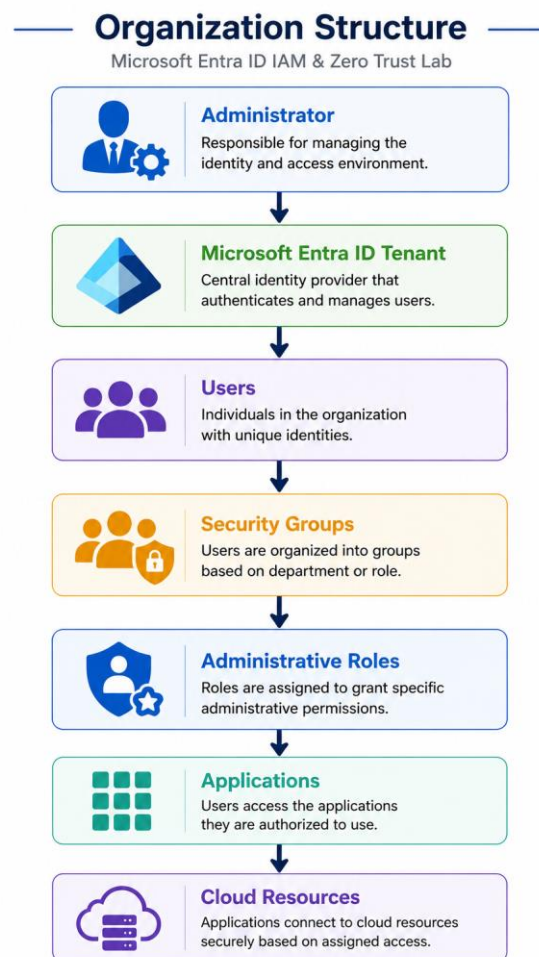
5. Technologies Used

Component	Technology
Cloud Identity Platform	Microsoft Entra ID
Portal	Microsoft Azure Portal
Identity Management	Microsoft Entra ID
Access Control	RBAC
Authentication	Username & Password
MFA	Microsoft Authenticator (Documented)

Component	Technology
User Management	Entra ID Users
Group Management	Security Groups
Documentation	Microsoft Word
Repository	GitHub

6. Lab Architecture

The project simulates a cloud organization called **CyberSecurity Lab**.



7. Implementation

The project was implemented in several phases.

Phase 1

Tenant Creation

A Microsoft Entra ID tenant was created using a Microsoft account.

Phase 2

User Creation

Six users were created.

Example users include:

- Muhammad Awais (Global Administrator)
- Umer (Manager / Helpdesk Administrator)
- Ahmed (Contractor)
- Asad (IT Employee)
- Ali (Employee)
- Bilal (HR Employee)

Each user was assigned unique identity information including:

- User Principal Name
- Display Name
- Department
- Employee ID
- Job Title
- Office Location

Phase 3

Security Groups

Department-based security groups were created.

Groups include:

- HR
- IT
- Managers

- Contractors

Each user was assigned to the appropriate group according to their department.

Phase 4

- **Administrative Roles:**

Administrative roles were assigned according to organizational responsibilities.

Examples include:

- 1. Global Administrator**

Responsible for complete tenant administration.

- 2. Helpdesk Administrator**

Responsible for password resets and user support.

- 3. Security Administrator**

Responsible for identity security management.

- 4. Standard Users**

Assigned no administrative privileges.

8. User Management

Microsoft Entra ID was used to manage user identities.

Operations performed include:

- Create users
- Modify user information
- Assign departments
- Assign job titles
- Assign security groups
- Assign administrative roles
- Delete users
- Simulate employee lifecycle

9. Security Groups

Department-based Security Groups were created to simplify access management.

Benefits include:

- Easier permission management

- Centralized administration
- Reduced administrative overhead
- Improved scalability

Groups implemented:

Group	Purpose
HR	Human Resources
IT	Information Technology
Managers	Department Managers
Contractors	External Users

10. Role-Based Access Control (RBAC)

Role-Based Access Control was implemented by assigning administrative roles instead of assigning permissions directly to users.

Assigned Roles

User	Role
Muhammad Awais	Global Administrator
Umer	Helpdesk Administrator
Asad	Security Administrator
Ahmed	Standard User
Bilal	Standard User

RBAC reduces administrative complexity while improving security.

11. Least Privilege Implementation

The Principle of Least Privilege ensures users receive only the permissions necessary to perform their work.

Implementation:

- Standard users received no administrative permissions.

- Administrative roles were limited to designated personnel.
- Group membership determined department access.
- Excessive permissions were avoided.

Benefits:

- Reduced attack surface.
- Protection against insider threats.
- Reduced accidental changes.
- Improved compliance.

12. Joiner-Mover-Leaver (JML) Workflow

The project documents the complete user lifecycle.

Joiner

- HR submits request.
- User account created.
- Department assigned.
- Security group assigned.
- Administrative role assigned if required.
- MFA registration (documented).
- User receives access.

Mover

- Department changes.
- Old group removed.
- New group assigned.
- Permissions updated.
- Administrative role reviewed.
- Access review completed.

Leaver

- User account disabled or deleted.
- Active sessions revoked.
- Group memberships removed.
- Administrative roles removed.
- Licenses removed.
- Offboarding documented.

13. Access Reviews

Regular access reviews help maintain secure environments.

The review process verifies:

- Active users
- Department memberships
- Administrative roles
- Unused accounts
- Excessive permissions

Review information includes:

- Reviewer
- Date
- User
- Role
- Decision
- Comments

14. Security Policies

The project includes comprehensive security policies covering:

- Password Policy
- MFA Policy
- Identity Policy

- Least Privilege Policy
- RBAC Policy
- JML Policy
- Access Review Policy
- Administrative Account Policy
- Security Group Policy
- Offboarding Policy

15. Results

The project successfully achieved the following:

- ✓ Microsoft Entra ID tenant deployed.
- ✓ Users created.
- ✓ Security Groups configured.
- ✓ Users assigned to groups.
- ✓ Administrative roles assigned.
- ✓ RBAC implemented.
- ✓ Least Privilege applied.
- ✓ JML workflow documented.
- ✓ Security policies developed.
- ✓ GitHub repository created with project documentation.

16. Challenges

Several challenges were encountered during implementation:

- Understanding Microsoft Entra ID licensing limitations.
- Conditional Access unavailable in the Free edition.
- Advanced MFA requiring Premium licensing.
- Learning Microsoft administrative roles.

- Designing a realistic organizational structure.

These challenges were addressed through documentation and careful planning.

17. Lessons Learned

This project provided practical experience with:

- Microsoft Entra ID
- Identity and Access Management
- Zero Trust Architecture
- RBAC
- Administrative Roles
- Security Groups
- User Lifecycle Management
- Access Reviews
- Security Documentation
- Cloud Identity Management

It also improved familiarity with Microsoft Azure administration and enterprise IAM concepts.

18. Future Enhancements

Future improvements include:

- Implement Microsoft Entra ID Premium P1/P2.
- Configure Conditional Access policies.
- Enforce Multi-Factor Authentication.
- Integrate Microsoft Defender for Identity.
- Connect Microsoft Sentinel for SIEM monitoring.
- Automate JML workflows using Microsoft Entra Lifecycle Workflows.
- Configure Privileged Identity Management (PIM).
- Integrate Single Sign-On (SSO) applications.

- Implement Identity Governance features.

19. Conclusion

This project successfully demonstrates the implementation of a cloud-based Identity and Access Management (IAM) laboratory using Microsoft Entra ID. The environment includes user provisioning, department-based security groups, administrative role assignments, Role-Based Access Control (RBAC), least privilege enforcement, and documented Joiner-Mover-Leaver (JML) workflows. These components reflect how organizations manage identities securely while following Zero Trust principles.

Although some advanced features, such as Conditional Access, enterprise Multi-Factor Authentication (MFA), and Privileged Identity Management (PIM), require Microsoft Entra ID Premium licensing and were therefore documented rather than fully implemented, the project establishes a strong foundation in cloud identity management. It demonstrates practical skills relevant to SOC Analyst, IAM, Azure Administrator, and Microsoft Security roles, while showcasing an understanding of identity governance, access control, and enterprise security best practices.